Derivatives Basics

Arun Kumar

Indian Institute of Technology Ropar

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Derivatives

Definition (Derivatives)

Derivatives are financial securities whose payoffs are explicitly given by the value of some other underlying asset.

Examples

Important examples of derivatives are forward contracts, futures contracts, options, swaps, swaptions.

- The derivatives market is often estimated at more than $1.2 quadrillion (1200 trillion).
- Some reports says that the derivatives market is more than 10 times the size of the total world GDP.

Forward Contracts

Forward contract on an asset is a contract to buy or sell a unit amount of the asset at a specific price and at a fixed time in the future.

Example

Consider a Forward contract to buy 100,000 kg of sugar at 80 Rs per kg on 15th July.
Some features of Forward contracts:

- The contract is between two parties.
- The asset involved in the forward contract is called the **underlying asset**.
- **Buyer** is said to be in **long position** of the Forward contract.
- **Seller** is said to be in **short position**.
- The agreed price of the asset in the contract is called the **delivery price**.
- The time of trading of the asset in the contract is called **maturity**.
- The market for immediate delivery of the asset is called the **Spot Market**.
- The price of the asset in the spot market is called **Spot Price**.

Forward Contract Contd...

Cash-flow from forward contract:

- Holding a long position in a forward contract with forward price F and spot price at T of the underlying asset S_T is given by $(0, S_T - F)$, here time points are $t = 0, T$.
- Similarly, cash flow of short position in the Forward contract is $(0, F - S_T)$.
- Note that zero value for entering into the forward contract is due to the fact that the payoff at maturity T of the forward contract can be negative also.

No Arbitrage Assumption

Definition (Arbitrage)

The simultaneous buying and selling of a security at two different prices in two different markets, resulting in profits without risk. Perfectly efficient markets present no arbitrage opportunities. Perfectly efficient markets seldom exist, but, arbitrage opportunities are often precluded because of transactions costs.

Example

As a hypothetical example, if Infosys stock trades at Rs 790 on the National Stock Exchange (NSE) and at Rs 800 on the Bombay Stock Exchange (BSE), an investor could guarantee a profit by purchasing the stock on the NSE and simultaneously selling the same amount of stock on the BSE.

Long Position vs a Short Position in Equity

Long Position: If an investor has long positions, it means that the investor has bought and owns those shares of stocks.

Short Position: If the investor has short positions, it means that the investor owes those stocks to someone, but does not actually own them yet. Sometime this is also called **short selling** which we mean "borrow the asset and sell it in the spot market to be returned later".

Example

Consider an investor who has sold 100 shares of Google without yet owning those shares is said to be short 100 shares. The short investor owes 100 shares at settlement and must fulfill the obligation by purchasing the shares in the market to deliver.

Some Assumptions

Our current assumptions of the model for the Market are the following:

- No arbitrage.
- There is an ideal bank with a spot rate curve s_t, $t = 1, 2, \cdots$.
- Among other securities, Market contains the underlying asset.
- Short selling of the underlying asset is allowed.

Spot Rates

- **Spot rate** for maturity t denoted by s_t is defined as the yield (per annum) on zero coupon bond with maturity t.
- Thus the term structure of interest rates corresponding to firms belonging to a category is given by Spot rate curve is given by $\{s_t, t = 1, 2, \cdots\}$ where t denote beginning of each period.
- For example, if periods are divided semi annually, then $t = 0, 1, \cdots$ corresponds to the time points today, 6 months, 12 months, 18 months etc.

Discount Rates and PV

- Discount rate d_t corresponding to the spot rate s_t is given by

$$d_t = \frac{1}{(1+s_t)^t} \quad \text{(annual compounding)}$$

- Note that Rs. 1 today becomes $(1+s_t)^t$ at the end of t years.
- Similarly,

$$d_t = \frac{1}{(1+s_t/2)^{2t}} \quad \text{(semi-annual compounding)}$$

- Moreover $d_t = e^{-s_t t}$ (continuous compounding)
- The present value (with respect to the spot rate curve $\{s_t, t = 0, 1, \cdots\}$) of the cash flow $(x_0, x_1, \cdots, x_n)$ is

$$PV = x_0 + d_1 x_1 + \cdots + x_n d_n.$$

Example: constructing the spot curve

Market Data :	6 months bill		Price 97.44
	1 year bill		Price 94.88
	18 months Bond	coupon 5%	Price 99.38
	2 year Bond	coupon 6%	Price 100.89

Semi-annual coupon payment and all have face value \$ 100 .

$$97.44 = \frac{100}{(1+s_{.5}/2)}, \quad s_{.5} = 5.25\%.$$

$$94.88 = \frac{100}{(1+s_1/2)^2}, \quad s_1 = 5.32\%.$$

$$99.38 = \frac{2.5}{(1+s_{.5}/2)} + \frac{2.5}{(1+s_1/2)^2} + \frac{102.5}{(1+s_{1.5}/2)^3}, \quad s_{1.5} = 5.44\%.$$

$$100.89 = \frac{3}{(1+s_{.5}/2)} + \frac{3}{(1+s_1/2)^2} + \frac{3}{(1+s_{1.5}/2)^3} + \frac{103}{(1+s_2/2)^4}, \quad s_2 = 5.53\%.$$

Relation between current price and forward price

Theorem

Let F denote the forward price, T maturity time, $d(0,T)$ the discount factor for the period $[0,T]$ and S_0 current spot price of the underlying asset. Also assume that there is no storage cost or dividend for the underlying asset. Then

$$F = \frac{S_0}{d(0,T)}.$$

Proof.

The proof is based on the "no arbitrage argument". Suppose

$$F > \frac{S_0}{d(0,T)}$$

- Borrow S_0 from Bank. [Cash flow $(S_0, -\frac{S_0}{d(0,T)})$].
- Buy unit asset from the spot market and store. [Cash flow $(-S_0, S_T)$]
- Short unit forward contract. [Cash flow $(0, F - S_T)$].

This creates an "arbitrage". This can be seen from the following observation. Initial value of the above investment is zero and it has a final payoff of $F - \frac{S_0}{d(0,T)}$, an arbitrage opportunity. $\square$

Forward rate in FX market: Covered Interest Rate Parity

The forward foreign exchange rate are given by

$$F = S\frac{(1+i_f)}{(1+i_d)},$$

where

- i_d =The interest rate in the domestic currency or the base currency
- i_f = The interest rate in the foreign currency or the quoted currency
- S = The current spot exchange rate
- F = The forward foreign exchange rate

Forward contract with storage cost

- Now we assume the underlying asset has carrying cost c_k for the period $[k, k+1), k = 0, 1, \cdots, M$ payable at the beginning of the period.
- M is the number of periods until maturity of the forward contract.
- If underlying is a commodity, then carrying cost (storage cost) can be cost to store or insurance fee etc. which in general positive.
- If the underlying is a financial security such as stock, carrying is cost is dividend and hence is negative.

Theorem

The forward price assuming carrying cost is given by

$$F = \frac{S_0}{d(0,M)} + \sum_{k=0}^{M-1} \frac{c_k}{d(k,M)}.$$

Proof.

Let

$$F < \frac{S_0}{d(0,M)} + \sum_{k=0}^{M-1} \frac{c_k}{d(k,M)}.$$

Then

- Short sell one unit of underlying asset and receive S_0 at time 0.
- Invest S_0 in Bank at time 0.
- Long one unit Forward contract at time 0.
- Receive c_k at the beginning of the period $[k, k+1)$ and invest in Bank.

Clearly net value of the above investment at 0 is 0. The value at time T is

$$\frac{S_0}{d(0,M)} + \frac{c_0}{d(0,M)} + \frac{c_1}{d(1,M)} + \cdots + \frac{c_{M-1}}{d(M-1,M)} - F > 0.$$

Therefore, the above investment strategy is an arbitrage. $\square$

Value of a Forward contract

- Forward price varies with time. i.e. it is a function of time.
- Let F_t denote the forward price at time t.
- We have the following question. Suppose you are holding a long forward contract at time 0 and current time is t. What is the value of your contract today? i.e. suppose you would like to transfer your contract to a second party, how much she will pay(+ or -) for it.
- The answer is: value of the contract, $f_t = (F_0 - F_t)d(t, T)$.

Proof

- Consider the portfolio of one long forward contract with delivery price F_t and a short forward contract with delivery price F_0 at time t.
- The cash flow for the long position is

$$(0, S_T - F_t).$$

- The cash flow from the short position is $(-f_t, F_0 - S_T)$
- The net cash flow from the portfolio is $(-f_t, F_0 - F_t)$
- The portfolio has a deterministic cash flow and hence

$$f_t = (F_0 - F_t)d(t, T).$$

Example

A 10 year 16% bond with face value Rs. 100 is currently selling for Rs. 92. A long forward contract on this bond that has a delivery date in 1 year and a delivery price of Rs. 94. The bond pays coupons semi annually with one due 6 months from now and another just before the maturity of the forward contract. The current interest rates (compounded semi annually) for 6 months and 1 year are 7% and 8% respectively. Find the current value of the forward contract.

Solution. We have

$$d(0,1/2) = \frac{1}{1.035} = 0.966; d(0,1) = \frac{1}{1.04^2} = 0.924; d(1/2;1) = \frac{d(0,1)}{d(0,1/2)} = 0.956.$$

Delivery price $K = 94$. Let F_0 denote the current forward rate. Then price of the long forward contract with delivery price K is $f_0 = (F_0 - K)d(0,1)$. Let us calculate F_0.

$$F_0 = \frac{92}{d(0,1)} + \frac{-8}{d(1/2,1)} - 8 = 99.57 - 8.37 - 8 = 83.2.$$

Hence $f_0 = (83.2 - 94) \times 0.924 = -9.98$.

Futures Contract

- Futures contract is like forward contract with the difference of being traded in an organized exchange.
- Futures contract is designed to eliminate the default risk associated with forward contract.
- It is achieved through the process called marking to the market.

Futures Contract Contd...

- While entering into the futures contract both parties need to deposit an amount with the broker called the margin account.
- At the end of each day, the net receipt will be marked. i.e. at the end of each day, the delivery price will be revised to the current delivery (futures price) and the net receipt from this process is added to the market.
- For example, at the end of day one, for the party holding the long position in the futures contract $F_1 - F_0$ will be deposited in her/his margin account, where F_0 is the futures price at the time of entry of the contract and F_1 is the futures price at the end of day one.
- when the margin value falls below a certain level, the contract holder need to pay money to makeup the margin account.

Futures Contract on S&P 500

Underlying	S&P 500 index
Where traded	Chicago Mercantile Exchange
Size	$250 × S&P 500 index
Months	March, June, September, December
Trading ends	Business day prior to determination of settlement price
Settlement	Cash-settled, based upon opening price of S&P 500 on third Friday of expiration month

Margins and Marking to Market

- Suppose the futures price is 1100 and you wish to acquire a \$2.2 million position in the S&P 500 index.
- The notional value of one contract is $\$250 \times 1100 = \$275,000$; this represents the amount you are agreeing to pay at expiration per futures contract.
- To go long \$2.2 million of the index, you would enter into \$2.2 million/\$0.275 million = 8 long futures contracts.
- Both buyers and sellers are required to post a security deposit with the broker to ensure that they can cover a specified loss on the position. This deposit, which can earn interest, is called **margin** and is intended to protect the counterparty against your failure to meet your obligations.

MoM Contd...

- suppose that there is 10% margin and daily settlement. The margin on futures contracts with a notional value of \$2.2 million is \$220,000.
- If the S&P 500 futures price drops by 1, to 1099, we lose \$2000 on our futures position.
- Thus, if the continuously compounded interest rate is 6%, our margin balance after 1 day is

$$220,000e^{0.06/365} - 2000 = 220036.17 - 2000 = 218036.17.$$

- The decline in the margin balance means the broker has significantly less protection should we default. For this reason, participants are required to maintain the margin at a minimum level, called the maintenance margin. This is often set at 70% to 80% of the initial margin level.
- The broker make a **Margin Call**.

Options

Definition (Option)

An option is a financial contract where the buyer of the contract has the right to buy/sell specific number of units of the underlying asset at a specified price on or before a Specified date. This right is granted by the seller of the option.

Terminologies related to options

- Buyer of the option is called holder and seller is called writer.
- The specific time till the option is valid is called the maturity.
- Specified price = **strike** or **exercise price**
- Buyer = holder = long position
- Seller = writer = short position
- If the holder has the write to buy, it is called **call option**.
- If the holder has the write to sell, it is called **put option**.
- On/before: **American**. Only on expiration: **European**.

In the Money and Out of the Money Options

- A call option with strike K is said the be **in the money option** if $S_0 > K$.
- Call option is called **out of the money** if $S_0 < K$.
- It is called **at the money** if $S_0 = K$.
- Here S_0 is the current price of the underlying asset.
- The payoff at maturity of a European call option is

$$(S_T - K)^+ = \max\{S_T - K, 0\},$$

where S_T is the price of the underlying at the maturity T.

TCS Option Chain

TCS OPTIONS CHAIN NIFTY

Expiry Dates (TCS) : 28APR2022

Expiry Dates (NIFTY) : --Select--

TCS Expiry Date 28APR2022 As on 02-FEB-2022 16:00:00 Hours IST

Calls											Puts									
Quote	Open Interest	Change in Open Interest	LTP	Net Change	Volume	Bid Qty	Bid Price	Offer Price	Offer Qty	Strike Price	Bid Qty	Bid Price	Offer Price	Offer Qty	Volume	Net Change	LTP	Open Interest	Change in Open Interest	Quote
Quote	-	-	-	-	-	-	-	-	-	3300.00	-	-	30.00	150	-	-	-	-	-	Quote
Quote	-	-	-	-	-	-	-	-	-	3400.00	-	-	30.00	150	-	-	-	-	-	Quote
Quote	-	-	-	-	-	-	-	-	-	3500.00	150	31.00	50.95	150	9	-8.00	37.00	1,050	900	Quote
Quote	-	-	-	-	-	-	-	-	-	3600.00	300	45.00	57.90	150	4	-111.20	55.15	300	300	Quote
Quote	-	-	-	-	-	-	-	-	-	3700.00	150	75.00	-	-	-	-	-	-	-	Quote
Quote	-	-	-	-	-	150	103.00	-	-	3800.00	150	121.05	129.65	150	12	-12.00	123.00	1,800	1,650	Quote
Quote	-	-	-	-	-	150	73.80	-	-	3900.00	150	127.00	-	-	-	-	-	-	-	Quote
Quote	9,750	2,550	135.00	12.45	23	150	132.45	135.55	150	4000.00	150	171.00	325.00	150	-	-	-	-	-	Quote
Quote	-	-	-	-	-	150	45.00	-	-	4060.00	-	-	-	-	-	-	-	-	-	Quote
Quote	150	-	-	-	-	150	89.00	100.55	150	4100.00	-	-	-	-	-	-	-	-	-	Quote
Quote	-	-	-	-	-	150	41.00	80.55	150	4200.00	-	-	-	-	-	-	-	-	-	Quote
Quote	-	-	-	-	-	150	30.00	100.00	150	4300.00	-	-	-	-	-	-	-	-	-	Quote
Quote	150	150	34.00	3.50	10	1,200	28.00	39.00	150	4400.00	-	-	-	-	-	-	-	-	-	Quote